

Assembly Overview

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(patchwork)
```

I'm aggregating the QUASt output of the assemblies of the different layers

1. Loading data

```
names=sort(list.files('./input/nf-coreMAG/Assembly/MEGAHIT/QC/'))
```

```
data_list <- lapply(names, function(file) {
  read.csv(paste0('./input/nf-coreMAG/Assembly/MEGAHIT/QC/',file,'/QUAST/report.tsv'), sep = "\t", header = 1)
})
```

```
data = Reduce(function(x, y) cbind(x, y), data_list) %>% mutate(TDM009 = 0) %>% t() %>% data.frame() %>%
```

```
# Fixing the column names
```

```
name_dict=c(
X..contigs.....0.bp.= "nb_contig_min_0_bp",
X..contigs.....1000.bp. = "nb_contigs_min_1000_bp",
X..contigs.....5000.bp. = "nb_contigs_min_5000_bp",
X..contigs.....10000.bp. = "nb_contigs_min_10000_bp",
X..contigs.....25000.bp. = "nb_contigs_min_25000_bp",
X..contigs.....50000.bp.= "nb_contigs_min_50000_bp",
Total.length.....0.bp.   = "Total_length_gt_0_bp",
Total.length.....1000.bp. = "Total_length_gt_1000_bp",
Total.length.....5000.bp. = "Total_length_gt_5000_bp",
Total.length.....10000.bp. = "Total_length_gt_10000_bp",
Total.length.....25000.bp. = "Total_length_gt_25000_bp",
Total.length.....50000.bp. = "Total_length_gt_50000_bp",
X..contigs = "nb_contigs",
Largest.contig = "Largest_contig",
Total.length = "Total_length",
GC.... = "GC_perc",
N50 = "N50",
N75 = "N75",
```

```

L50           = "L50",
L75           = "L75",
X..N.s.per.100.kbp = "nb_Ns_per_100_kbp")

colnames(data)[colnames(data) %in% names(name_dict)] <-
  name_dict[colnames(data)[colnames(data) %in% names(name_dict)]]

write.table(data, './output/assembly_stats.csv', quote = F, row.names = F)

summary(data[10:70,])

##  nb_contig_min_0_bp nb_contigs_min_1000_bp nb_contigs_min_5000_bp
##  Min.      :    0      Min.      :    0      Min.      :    0.0
##  1st Qu.: 24576      1st Qu.: 4754      1st Qu.: 342.0
##  Median : 32102      Median : 6372      Median : 571.0
##  Mean   : 36892      Mean    : 7950      Mean    : 746.7
##  3rd Qu.: 49128      3rd Qu.: 9462      3rd Qu.:1088.0
##  Max.    :101740      Max.     :25203      Max.     :2535.0
##  nb_contigs_min_10000_bp nb_contigs_min_25000_bp nb_contigs_min_50000_bp
##  Min.      : 0.0      Min.      : 0.00      Min.      : 0.000
##  1st Qu.: 53.0      1st Qu.: 4.00      1st Qu.: 0.000
##  Median :158.0      Median : 15.00      Median : 1.000
##  Mean   :209.9      Mean    : 26.05      Mean    : 3.066
##  3rd Qu.:295.0      3rd Qu.: 44.00      3rd Qu.: 4.000
##  Max.    :711.0      Max.     :113.00      Max.     :27.000
##  Total_length_gt_0_bp Total_length_gt_1000_bp Total_length_gt_5000_bp
##  Min.      :    0      Min.      :    0      Min.      :    0
##  1st Qu.: 22218832      1st Qu.:12467167      1st Qu.: 3105985
##  Median : 34180250      Median :17030838      Median : 5375124
##  Mean   : 35248760      Mean    :20932041      Mean    : 7239779
##  3rd Qu.: 44804402      3rd Qu.:27981500      3rd Qu.:10184403
##  Max.    :106107916      Max.     :67138443      Max.     :23358960
##  Total_length_gt_10000_bp Total_length_gt_25000_bp Total_length_gt_50000_bp
##  Min.      :    0      Min.      :    0      Min.      :    0
##  1st Qu.: 857893      1st Qu.: 162250      1st Qu.:    0
##  Median : 2635720      Median : 500117      Median : 60971
##  Mean   : 3601534      Mean    : 969439      Mean    : 212865
##  3rd Qu.: 5848618      3rd Qu.:1707181      3rd Qu.: 264698
##  Max.    :12280131      Max.     :5014467      Max.     :2025640
##  nb_contigs Largest_contig Total_length GC_perc
##  Min.      :    0      Min.      :    0      Min.      :    0      Min.      : 0.00
##  1st Qu.:11882      1st Qu.: 42037      1st Qu.:17929054      1st Qu.:60.21
##  Median :16722      Median : 58196      Median :25519613      Median :61.29
##  Mean   :19019      Mean    : 68796      Mean    :28535503      Mean    :60.40
##  3rd Qu.:25317      3rd Qu.: 79957      3rd Qu.:36525884      3rd Qu.:62.67
##  Max.    :56945      Max.     :285743      Max.     :88963704      Max.     :65.67
##  N50      N75      L50      L75
##  Min.      :    0      Min.      : 0.0      Min.      :    0      Min.      :    0
##  1st Qu.:1437      1st Qu.: 816.0      1st Qu.:1877      1st Qu.: 5484
##  Median :1884      Median : 937.0      Median :2821      Median : 6917
##  Mean   :2067      Mean    : 957.3      Mean    :3364      Mean    : 8606
##  3rd Qu.:2419      3rd Qu.:1074.0      3rd Qu.:4953      3rd Qu.:12433
##  Max.    :5232      Max.     :1635.0      Max.     :9307      Max.     :24790
##  nb_Ns_per_100_kbp

```

```
## Min.      :0
## 1st Qu.:0
## Median :0
## Mean     :0
## 3rd Qu.:0
## Max.      :0
```

2. Plots

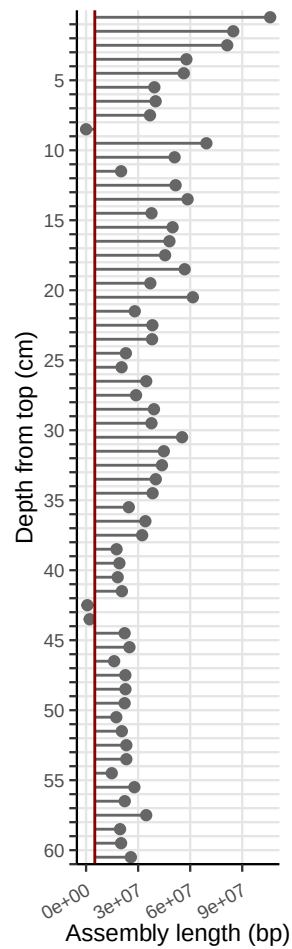
```
data_long <- data %>%
  mutate(sample=rownames(data)) %>%
  mutate(sample = str_remove(sample, "^MEGAHIT-group-")) %>%
  mutate(category=(case_when(
    str_starts(sample, "EXB") ~ "EXB",
    str_starts(sample, "TDM") ~ "TDM",
    TRUE ~ NA_character_ # for any other cases
  ))) %>%
  pivot_longer(cols= -c(sample, category))

desired_order=paste0(rep('TDM0', times=61), c('01','02','03','04','05','06', '07', '08', '09', as.character(1:5)))

data_long$depth_cm <- as.numeric(sub("TDM0*", "", data_long$sample))

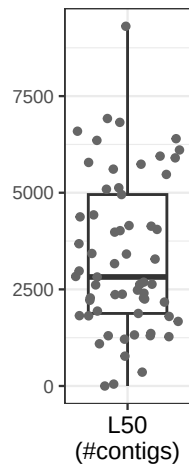
## Warning: NAs introduced by coercion

p1= data_long[data_long$name %in% c( "Total_length_gt_0_bp" ) & data_long$category=='TDM',] %>%
  ggplot(aes(y=depth_cm-0.5, x=value))+
  geom_segment( aes(y=depth_cm-0.5, yend=depth_cm-0.5, x=data_long[data_long$name %in% c( "Total_length_gt_0_bp" ) & data_long$category=='TDM',]$value)) +
  geom_point( color="grey40", size=1.5)+
  geom_vline(xintercept = data_long[data_long$name %in% c( "Total_length_gt_0_bp" ) & data_long$category=='TDM',]$value) +
  scale_y_reverse( breaks = seq(1, 61, 1), labels=ifelse(1:61 %% 5 == 0, 1:61, ""), limits = c(61,0), expand=c(0,0,0,0)) +
  theme_classic(base_size = 9)+
  theme(axis.text.x = element_text(angle = 30, vjust = 0.8, hjust = 1),
        axis.ticks.length = unit(0.1, "cm"),
        panel.grid.major = element_line(color = "grey90")) +
  ylab('Depth from top (cm)')+
  xlab('Assembly length (bp)')
p1
```



```
ggsave(' ../../06_figures/Fig1/E.pdf', height = 5, width=1.5, units = 'in')
```

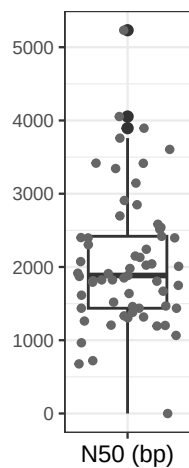
```
p2=data_long[data_long$name %in% c('L50') & data_long$category=='TDM',] %>% ggplot(aes(x=category, y=va
  geom_boxplot()+
  geom_jitter(width=0.5, color='gray40', size=1)+
  theme_bw(base_size = 9)+
  xlab('L50\n(#contigs)')+
  ylab('')+
  theme(axis.text.x = element_blank())
p2
```



```
ggsave('.../06_figures/Fig1/F.pdf', height=2.5, width=1.2, units = 'in')
saveRDS(p2, '.../06_figures/Fig1/F.rds')
```

```
p3=data_long[data_long$name %in% c('N50') & data_long$category=='TDM',] %>% ggplot(aes(x=category, y=value)) +
  geom_boxplot() +
  geom_jitter(width=0.5, color='gray40', size=1) +
  theme_bw(base_size = 9) +
  xlab('N50 (bp)') +
  ylab('') +
  theme(axis.text.x = element_blank())
```

p3



```
ggsave('.../06_figures/Fig1/G.pdf', height=2.5, width=1.2, units = 'in')
saveRDS(p3, '.../06_figures/Fig1/G.rds')
```